

VAL' KARKAY, Russian Federation

WMO: 219780

Lat: 70.084N

Lon: 170.925E

Elev: 6

StdP: 101.26

Time Zone: 12.00 (E12)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -38.5	(c) -36.9	(d) -40.5	(e) 0.1	(f) -38.3	(g) -39.0	(h) 0.1	(i) -36.7	(j) 22.3	(k) -13.8	(l) 19.4	(m) -10.1	(n) 2.2	(o) 210	(p) 1.095

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.1	18.1	12.4	14.9	10.6	11.6	8.7	12.8	17.5	10.8	14.7	8.8	11.5	10.0	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.9	7.6	14.7	8.1	6.7	12.6	6.4	6.0	10.1	36.1	17.8	31.3	14.7	26.5	11.6	16.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 15.9	(b) 13.5	(c) 12.0	(d) DB	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A	(q) N/A
(a) 15.9	(b) 13.5	(c) 12.0	(d) WB	(e) -38.7	(f) 14.0	(g) 2.0	(h) 2.7	(i) -40.1	(j) 15.9	(k) -41.3	(l) 17.5	(m) -42.4	(n) 19.0	(o) -43.9	(p) 20.9	(q) N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.8	-24.7	-23.9	-23.3	-13.5	-5.0	2.3	5.0	4.2	1.7	-5.5	-13.7	-22.1
	DBStd	12.73	7.11	7.80	7.15	6.09	4.49	3.61	3.98	3.65	3.31	5.71	7.53	8.30
	HDD10.0	7254	1076	948	1033	704	464	235	169	185	253	481	712	994
	HDD18.3	10271	1334	1182	1291	954	722	481	415	437	501	739	962	1253
	CDD10.0	26	0	0	0	0	0	4	13	7	2	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
	WSAvg	5.2	5.6	5.3	4.3	4.5	4.8	5.2	5.5	5.7	5.5	4.9	5.7	5.6
	PrecAvg	214	10	10	9	8	9	17	27	36	32	27	19	12
(15) Precipitation	PrecMax	285	21	23	25	20	26	38	47	61	69	83	41	26
	PrecMin	158	3	1	0	2	1	5	11	18	10	7	8	2
	PrecStd	35	4	5	6	4	4	7	10	13	13	15	8	7
	PrecStd	35	4	5	6	4	4	7	10	13	13	15	8	7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.6	0.7	1.8	2.9	5.6	20.5	22.4	18.8	16.4	6.8	2.3	1.5
		MCWB	-0.8	-0.3	0.3	0.6	1.8	12.7	14.8	12.5	12.0	4.8	0.5	-0.1
	2%	DB	-2.0	-4.6	-3.9	0.8	3.6	16.2	19.1	15.4	11.6	3.6	-0.1	-4.0
		MCWB	-3.7	-5.6	-4.8	-1.1	1.5	10.5	13.2	11.4	8.6	2.6	-1.1	-4.7
	5%	DB	-11.8	-9.5	-9.9	-1.4	2.1	11.8	15.7	12.8	8.0	2.6	-1.8	-7.0
		MCWB	-12.3	-9.9	-10.5	-2.8	0.6	7.9	11.5	9.6	5.7	2.0	-2.7	-7.5
	10%	DB	-16.6	-13.8	-13.1	-4.0	0.8	7.4	11.5	10.2	5.6	1.2	-3.8	-10.0
		MCWB	-16.8	-14.0	-13.4	-4.7	0.0	4.9	8.9	8.4	4.4	0.7	-4.6	-10.4
	0.4%	WB	-0.6	-0.3	0.1	0.5	2.7	12.8	15.2	13.2	11.9	5.0	0.5	0.1
		MCDB	0.7	0.7	1.8	2.6	4.3	20.1	21.1	16.6	15.4	6.2	1.8	1.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	-3.5	-5.7	-4.8	-0.9	1.6	10.5	13.7	11.5	8.7	3.0	-0.8	-4.9
		MCDB	-2.2	-4.6	-3.7	0.7	3.0	16.1	18.5	15.1	11.6	3.3	0.0	-3.9
	5%	WB	-12.0	-9.9	-10.5	-2.7	0.9	7.7	11.4	9.9	6.1	1.9	-2.5	-7.5
		MCDB	-11.7	-9.4	-10.0	-1.5	2.0	11.6	15.9	12.3	7.6	2.4	-1.8	-7.0
	10%	WB	-16.8	-14.0	-13.4	-4.8	0.1	5.0	8.8	8.2	4.6	0.7	-4.6	-10.4
		MCDB	-16.6	-13.8	-13.0	-3.9	0.9	7.2	11.4	10.1	5.5	1.2	-3.8	-9.9
	5% DB	MDBR	5.4	5.3	6.5	7.1	4.7	4.6	5.1	4.2	3.4	4.2	5.5	5.0
		MCDBR	11.4	9.4	9.6	7.7	6.0	12.6	12.7	10.3	7.6	4.1	5.8	7.1
		MCWBR	10.6	8.6	9.0	6.7	4.7	7.8	7.9	6.8	5.0	3.4	5.2	6.6
		MCWBR	11.5	8.9	9.7	7.4	4.8	12.2	12.1	9.4	7.1	3.5	5.9	7.0
(38) Clear-Sky Solar Irradiance	taub	N/A	0.207	0.283	0.324	0.349	0.346	0.370	0.373	0.311	0.245	N/A	N/A	N/A
		taud	N/A	2.016	2.027	1.974	2.025	2.259	2.334	2.337	2.460	2.124	N/A	N/A
	Ebn at Noon	0	527	697	785	816	840	803	747	708	451	0	0	0
		Edh at Noon	0	39	85	124	135	111	100	89	58	35	0	0
(44) All-Sky Solar Radiation	RadAvg	0.00	0.33	1.96	4.36	6.08	6.51	5.16	3.28	1.63	0.46	0.00	0.00	0.00
	RadStd	0.00	0.05	0.11	0.20	0.34	0.52	0.41	0.30	0.12	0.06	0.00	0.00	0.00

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page